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Idea Submission · Organised by the Indian Space Research Organisation (ISRO)

Problem Statement **SIH26173**

Problem Statement ID	SIH26173
Problem Statement Title	iTantra — Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links
Theme	Smart Automation
PS Category	Software
Team ID	[PORTAL TEAM ID]
Team Name	B.J.P (Bug Janta Party)

iTantra

Speech in. Text across the link. Speech out.

An offline Android app that recognises speech on the sending phone, transmits only compact text over Bluetooth or Wi-Fi Direct, and speaks it aloud on the receiving phone.

Fully offline **10 languages**

No cloud speech API · On-device processing · Text-only communication

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iTantra — send voice as text, not as audio

IDEA TITLE · Proposed Solution

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What crosses the link

STEP 1 **VOICE** Mic, 16 kHz → STEP 2 **TEXT** ASR on device → STEP 3 **TRANSMIT** 49–111 bytes → STEP 4 **VOICE** TTS on device

Same sentence, two ways

Raw PCM audio, 3 s		≈ 93 KB
iTantra text payload		49–111 B

Measured payload from 4 test utterances. Roughly 1,000× smaller than the equivalent audio.

Proposed solution

An offline Android app: speech in, text out, a compact packet over Bluetooth or Wi-Fi Direct, speech back out on the receiving phone.

Detailed explanation

Push-to-talk works like a walkie-talkie. When idle, both phones run only a voice-activity detector. The speaker's phone carries the ASR cost; the listener's phone carries the TTS cost.

How it addresses the problem

Audio is too large for low-bitrate links. One spoken sentence becomes 49–111 bytes of UTF-8 text, so voice fits where it could not before.

Innovation and uniqueness

An asymmetric pipeline — one heavy model per phone. Only 0.6 MB of VAD runs while listening, so idle CPU stays near zero by design.

TECHNICAL APPROACH

Local intelligence, text-only transport

Technologies to be used

programming languages, frameworks, hardware

Android app

Kotlin · minSdk 21 (Android 5.0) · targetSdk 33 · native

AudioRecord / AudioTrack / BluetoothSocket / WifiP2pManager

ASR

Meta Omnilingual 300M CTC INT8 · 348.4 MB · Apache 2.0

TTS

Piper VITS · hi_IN pratham 60.6 MB + en_US lessac 60.3 MB · MIT

VAD

Silero · 0.6 MB · MIT · RTF 0.004 (≈240× faster than real time)

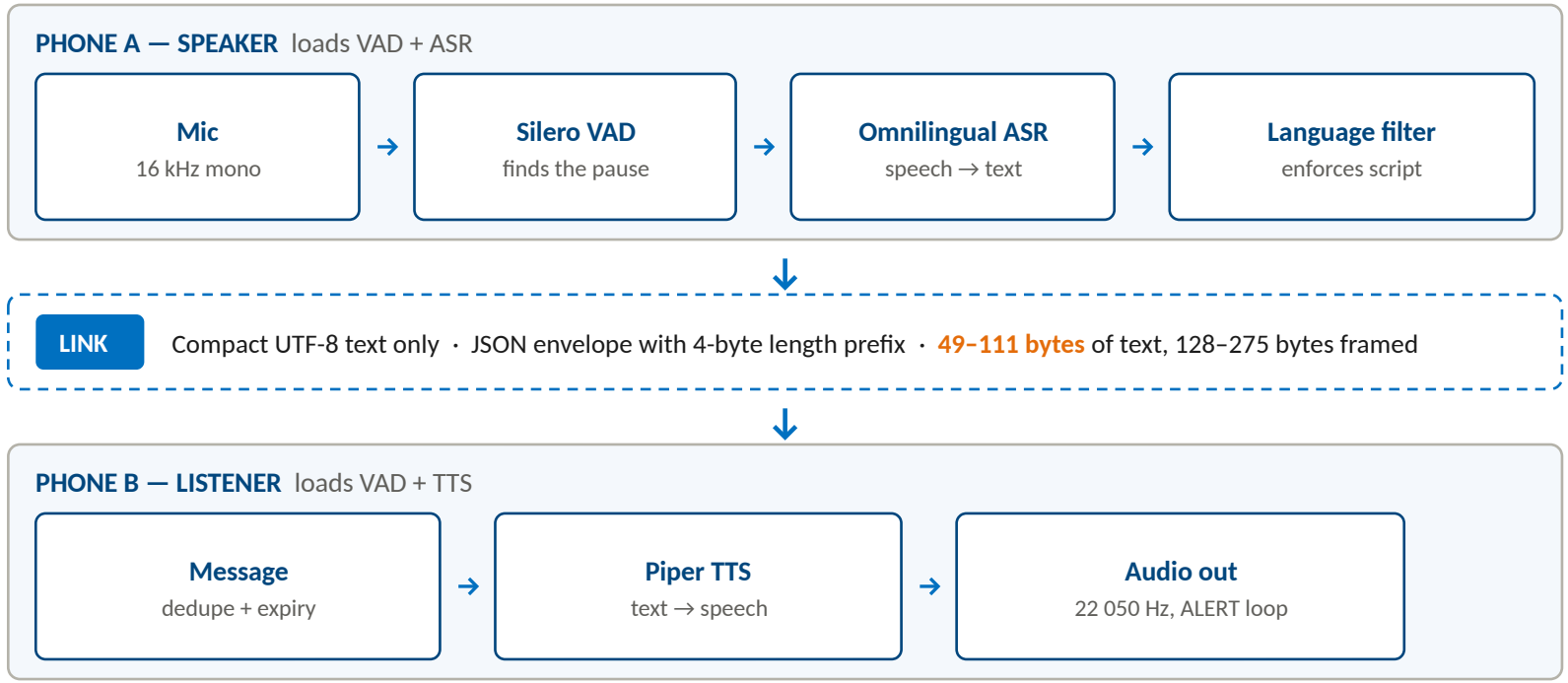
Runtime

sherpa-onnx v1.13.8 · Apache 2.0 · arm64-v8a

Transport

Bluetooth Classic RFCOMM (primary) · Wi-Fi Direct (fallback)

Methodology and process for implementation



One heavy model per phone. The speaker loads VAD + ASR; the listener loads VAD + TTS. A naive design loads both models on both phones and doubles the memory for no benefit. On Android the speaker measures **428 MB PSS**; the listener needs roughly 100 MB.

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FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea · measured, not projected

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Offline ASR on real human speech — Google FLEURS

lower is better

Language	WER (norm)	CER (norm)	RTF median
Kannada	0.307	0.053	0.041
Malayalam	0.361	0.059	0.041
English	0.181	0.067	0.040
Gujarati	0.300	0.071	0.041
Bengali	0.359	0.100	0.040
Telugu	0.431	0.100	0.039
Marathi	0.416	0.108	0.040
Hindi	0.281	0.113	0.040
Tamil	0.501	0.129	0.040
Odia	0.509	0.134	0.043

Model v2 (ctc-v2-int8) · 200 clips per language, 2,000 clips, 6.72 h · mean CER 0.093 · every language under 0.14. Normalisation strips punctuation and case because the CTC decoder emits neither; raw WER runs ≈1.3× higher.

Fast local processing

RTF median 0.039–0.043 across all ten. A 3-second sentence costs ≈0.12 s of compute.

Full multilingual coverage

All ten PS-mandated languages measured. Best Kannada 0.053, weakest Odia 0.134.

Against a far larger model

Meta's 7B LLM-ASR is 2–3× better — but it is 25 GiB and cannot run on a phone. Ours is 348 MB and does. Malayalam within 1.1×, Telugu 1.5×.

Challenge / Risk		Engineering fix / Strategy
The radio link has never connected — RFCOMM socket and Wi-Fi Direct group formation are unexecuted.	→	Next milestone: two physical phones. Adapter access and discovery already verified.
Every figure is from an API 34 emulator, not a low-end handset.	→	Benchmark on the cheapest available phone first; ship a release build.
Hindi degrades sharply in noise — CER 0.133 → 0.424 at 30 dB SNR.	→	Report CER beside WER; add a noise-robust front end; scope the demo to Hindi + English.
Kannada, Tamil and Odia TTS voices are not yet wired.	→	Route through espeak-ng; ship one voice per language to cut APK size.

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IMPACT AND BENEFITS

The network carries text. The device handles the voice.

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Potential impact on the target audience

Field responders

Communicate over constrained or congested local links, with no infrastructure to depend on.

Disaster / shelter teams

Useful exactly when reliable internet and cellular networks are unavailable.

Remote / industrial sites

Short voice communication where the link is narrow and bandwidth is at a premium.

Benefits of the solution

Social

Voice in, voice out. Usable by people who cannot read or type, because alerts are heard rather than read.

Inclusive by construction, not by add-on.

Economic

No paid API, no cloud subscription, no proprietary SDK. It runs on phones people already own.

Every model is Apache 2.0 or MIT — no licence fee.

Environmental

Sending text instead of audio lowers the transmitted energy per message by roughly three orders of magnitude.

Battery runtime effect still to be measured.

What is measured and what is not. Measured: accuracy on 2,000 real-speech clips across all ten languages, Android ASR and TTS latency, and memory with the model resident. Not yet measured: field accuracy in noise, phone-class latency, human TTS legibility, and battery runtime. The benefits above are **proposed outcomes, not measured results** — we claim no lives saved.

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RESEARCH AND REFERENCES

Details / Links of the reference and research work

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Models and runtime

ASR model	csukuangfj/sherpa-onnx-omnilingual-asr-1600-languages-300M-ctc-int8-2025-11-12 Apache 2.0 · 348.4 MB · INT8
TTS voices	rhasspy/piper-voices — hi_IN pratham, en_US lessac MIT · 60.6 MB + 60.3 MB · VITS
Runtime	k2-fsa/sherpa-onnx v1.13.8 + sherpa-onnx-v1.13.8-android native libs Apache 2.0 · arm64-v8a
Voice activity detection	Silero VAD — pause and boundary detection MIT · 0.6 MB · RTF 0.004

Evaluation data and platform

Speech corpus	google/fleurs — real recorded human speech, all ten PS-mandated languages CC-BY-4.0 · 200 clips per language, 2,000 clips, 6.72 h · evaluation only, not shipped
Transport docs	developer.android.com — Bluetooth RFCOMM and Wi-Fi Direct guides Platform documentation
Audio APIs	AudioRecord 16 kHz · AudioTrack 22 050 Hz · AudioManager STREAM_ALARM Android platform, API 21+
Reproducibility	fleurs_eval_full.py · tts_intelligibility.py · measure_ram_all.py · end_to_end.py Every figure in this deck is reproducible from these scripts

Licensing, component by component

ASR model **Apache 2.0** · Piper voices **MIT** · Silero VAD **MIT** · sherpa-onnx runtime **Apache 2.0** · AndroidX / Material **Apache 2.0** · espeak-ng-data **GPL (phoneme dictionaries)** — redistribution obligation flagged for review.

No CC-BY-NC component ships in the product; MMS-TTS was evaluated and rejected on that basis. Pairing is deliberately manual, so the sender of an emergency alert is authenticated.